

Workshop: The simplest sine wave generator.

Now that you know all about electromagnetic waves, sinusoidal waves and how wireless communication works, it is time for you to make your own sine wave generator.

Components required

- An NPN transistor. This is one of the basic building blocks in modern electronics.
- Capacitors. As standard charge storage devices, capacitors are available just as easily as transistors. You will need two 30 microfarad capacitors and a 10 to 100 microfarad variable capacitor box for this project.
- Two 10 ohm resistors. One 1 ohm resistor.
- Connecting wire.

Principle

A sine wave generator is the fundamental component of most transmitter devices. The circuit, known as an oscillator, gives a sinusoidally varying output of a specified frequency and amplitude that can be modified by other circuits to carry meaningful information.

A sine wave generator, otherwise known as an oscillator, consists of three subcomponents. These are:

1. The frequency determining device: this is the device in which when a charge is induced, the charge oscillates back and forth, through simple harmonic motions, and creates a sine wave. However, this circuit alone is not enough as the energy loss inherent to electric circuits will ensure that this oscillating signal will soon die down.

Such a circuit is usually realised in practice by connecting a capacitor and an inductor together. The charge oscillates back and forth, while the energy keeps getting converted back and forth

